Hybrid Temporal Forecaster: Phase-1 Report

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***Abstract*—This report presents Phase-1 of a hybrid temporal forecasting system for electricity demand prediction. The focus is on exploratory data analysis, statistical baseline modeling, and understanding limitations of classical approaches. The dataset exhibits strong seasonal behavior and volatility, motivating the need for advanced forecasting techniques.**

I. Introduction

Accurate short-term forecasting is critical in domains such as energy systems and financial markets. Electricity demand data exhibits complex characteristics including seasonality, noise, and regime shifts. This project aims to build a hybrid forecasting pipeline combining statistical and machine learning approaches.

Phase-1 focuses on understanding the dataset and implementing a statistical baseline using ARIMA.

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II. Literature Review

Time series forecasting has traditionally relied on the Box-Jenkins methodology, which introduced the ARIMA model for capturing linear temporal dependencies.

Hyndman and Khandakar (2008) proposed automated methods for selecting optimal ARIMA parameters, reducing manual tuning efforts.

De Livera et al. (2011) introduced TBATS models capable of handling complex seasonal structures, including multiple seasonalities.

The M4 Competition (Makridakis et al., 2018) demonstrated that hybrid models outperform individual forecasting techniques across diverse datasets.

These studies highlight that while classical statistical models provide strong baselines, they are insufficient for complex real-world data, motivating hybrid approaches.

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III. Dataset Description

The dataset used is the PJME hourly electricity consumption dataset.

- Frequency: Hourly
- Duration: 2002–2018
- Target Variable: PJME_MW

The dataset contains real-world electricity demand with strong seasonal and volatile patterns.

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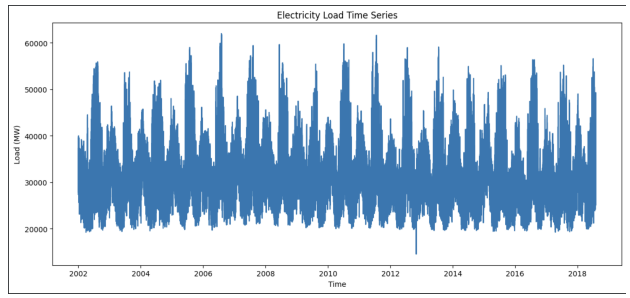

Fig. 1. Electricity Load Time Series

IV. Exploratory Data Analysis

A. Time Series Visualization

The time series shows strong periodic behavior with repeated peaks and troughs, indicating seasonality.

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B. Seasonality Analysis

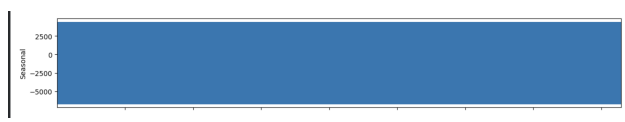

Fig. 2. Seasonal Patterns

Clear daily and weekly patterns are observed, confirming strong seasonality.

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C. Rolling Statistics

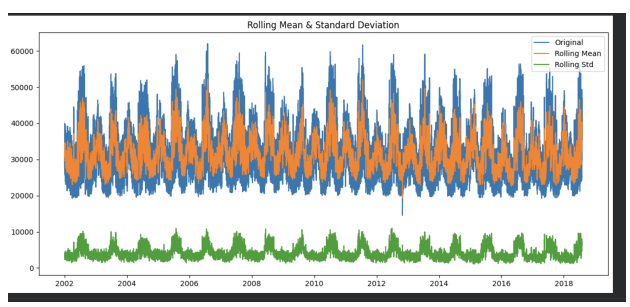

Fig. 3. Rolling Mean and Standard Deviation

The rolling mean and variance are not constant, indicating non-stationarity.

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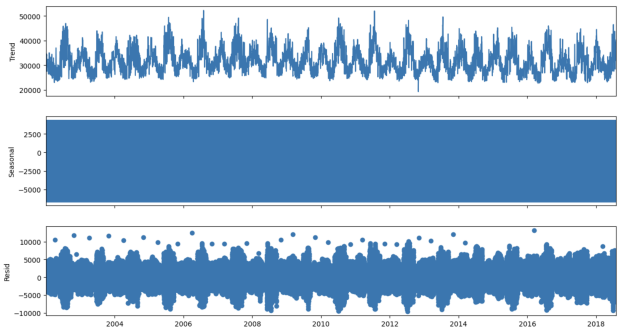

Fig. 4. Time Series Decomposition

D. Decomposition

The decomposition shows:

- Weak trend
- Strong seasonal component
- Significant residual noise

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V. Stationarity Analysis using ADF and KPSS Tests

To assess the stationarity of the time series, both the Augmented Dickey-Fuller (ADF) test and the KPSS test were applied.

A. ADF Test Results

- ADF Statistic: -20.0044
- p-value: 0.0

Since the p-value is less than 0.05, the ADF test suggests that the series is stationary.

B. KPSS Test Results

- KPSS Statistic: 1.3055
- p-value: 0.01

Since the p-value is less than 0.05, the KPSS test indicates that the series is non-stationary.

C. Combined Interpretation

The results of the two tests are contradictory:

- ADF test suggests stationarity (no unit root present)
- KPSS test suggests non-stationarity (instability in mean or variance)

This indicates that the time series is not strictly stationary. Instead, it is likely **trend-stationary but seasonally non-stationary**. The ADF test only checks for unit-root non-stationarity and does not account for strong seasonal patterns, whereas the KPSS test is sensitive to such structural variations.

D. Conclusion

The dataset exhibits **seasonal non-stationarity**, which is consistent with earlier observations from rolling statistics and time series decomposition.

VI. Baseline Model: ARIMA

An ARIMA model was implemented as a baseline using automatically selected parameters.

A. Model Performance

- MAE: 4986.0268461709575
- RMSE: 6498.279763683863

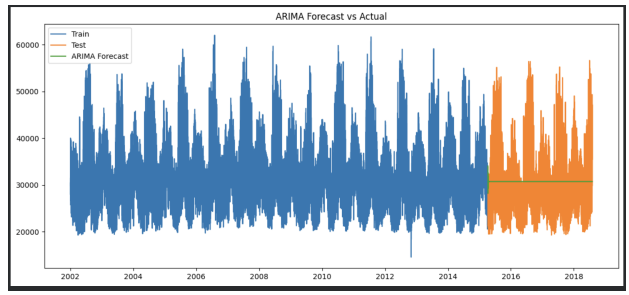

Fig. 5. ARIMA Forecast vs Actual

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VII. Discussion

The ARIMA model fails to capture the strong seasonal patterns present in the data, resulting in poor forecasting performance.

This is because ARIMA models only capture short-term linear dependencies and do not explicitly model seasonality.

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VIII. Why SARIMA is Needed

To address the limitations of ARIMA, Seasonal ARIMA (SARIMA) extends the model by incorporating seasonal components.

SARIMA is capable of capturing periodic patterns such as daily and weekly cycles in electricity demand data, making it more suitable for this problem.

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IX. Phase-2 Implementation Plan

In Phase-2, the following improvements will be implemented:

- Feature Engineering:
  - Lag features (lag_1, lag_24, lag_168)
  - Rolling statistics
- Machine Learning Models:
  - Random Forest
  - XGBoost
- Evaluation:
  - Comparison with ARIMA baseline

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X. Conclusion

This phase established a statistical baseline using ARIMA and conducted a comprehensive analysis of the dataset. The results highlight the limitations of classical models in capturing seasonal patterns, motivating the use of advanced and hybrid approaches in subsequent phases.

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XI. References

References

[1] G. Box and G. Jenkins, *Time Series Analysis*, 1970.

[2] R. Hyndman and Y. Khandakar, "Automatic time series forecasting," 2008.

[3] A. De Livera et al., "Forecasting time series with complex seasonality," 2011.

[4] S. Makridakis et al., "The M4 Competition," 2018.